Dumbledore Solution
Technology for Transparency and Social Impact

IMPLEMENTATION EVIDENCE

Kano State Health Finance Tracker

Implementation Report

AGREEMENT	Dumbledore-2026C-SPARK-NGA
CLIENT CONTEXT	International Budget Partnership / Kano State health financing implementation
CONTRACT DELIVERY DATE	31 July 2026
REPORT DATE	6 August 2026
PRODUCTION SYSTEM	https://budget-tracker--budget-tracker-2fb26.us-east4.hosted.app
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Prepared for implementation review and contractual closeout

1. Executive Summary

Implementation outcome

Dumbledore Solution has delivered a production-operational health finance accountability platform that replaces the source workbook's ledger, lookup, budget, AOP, dashboard, and analytical functions with a secure database-backed web application.

The system is deployed on Firebase App Hosting and connected to Supabase/Postgres for authenticated, role-scoped data access.

Status summary

Assessment	Count	Meaning
Completed	7	Implemented and evidenced in code, documentation, live data, deployment, or the supplied completion marks.
Substantial	5	Core capability is operating; extended automation, a formal report, or client acceptance remains.
Partial	1	Development and production exist; a dedicated staging environment is not evidenced.

2. Contract Deliverables and Implementation Status

Status is based on observable implementation evidence. **Substantial** means the primary capability is delivered, but part of the precise contractual wording still requires completion, configuration, or formal acceptance.

No.	Contract deliverable	Status	Implementation evidence and assessment
1	Workbook Review Report	**Completed**	Workbook structure, formulas, validation lists, ledger logic, budget, AOP, facility/LGA relationships, and workbook defects are documented in `docs/workbook-study.md`. Q2 source comparisons and the health-component review PDF add implementation-period reconciliation evidence.
2	Technical Architecture Document	**Completed**	Architecture is documented through ADRs, `database-model.md`, `supabase-implementation.md`, Next.js server conventions, and implementation plans. The delivered stack is Next.js, Supabase/Postgres, Firebase App Hosting, role-based access, and offline client storage.
3	Infrastructure Audit Report	**Completed**	The supplied contract excerpt bears a completion mark. Deployment readiness is evidenced by a successful production build, Firebase rollout, HTTPS endpoint, managed Supabase database, environment-variable controls, and documented infrastructure gaps. This consolidated report records the remaining infrastructure closeout items.
4	Data Model and Field Specification	**Completed**	Field-by-field workbook-to-web mapping and a normalized table specification are documented. Sixteen SQL migrations implement foreign keys, checks, triggers, RLS, audit history, publication locks, facilities, approved budget lines, revenues, and monthly tracking.
5	All Three Environments Configured	**Partial**	Local development and Firebase production are evidenced. A separate staging project/environment and a documented three-environment release path were not found in the checked repository or deployment configuration.
6	Backend API and Database	**Completed**	Supabase/Postgres is populated with reference, budget, AOP, facility, ledger, revenue, and monthly-tracking data. Next.js route handlers cover account administration, invitations, email workflow, and assistant chat. Nine reporting views and application aggregators support dashboards and exports.
7	All 21 MDA Dashboards	**Completed**	The MDA dashboard is dynamically scoped by authenticated membership and uses the same reporting engine for all 21 budgeted health MDAs. It includes funding, expenditure, budget execution, programme, PHC, AOP, filters, validation, offline caching, and queued synchronization.
8	SMoH Dashboard	**Substantial**	Statewide Admin Insights is implemented with overview, MDA, programme, PHC, and AOP tabs; monthly reporting coverage; component reconciliation; watchlists; BIR, audit, CSO, and budget/planning reports; and PDF/CSV exports. A full upload-based BIR validation workflow and production AI service.
9	MoPB/MoF Aggregate Feed	**Substantial**	A dedicated Budget and Planning report computes pro-rata execution, zero-spend MDAs, AOP variance, and planning snapshots. Monthly MDA tracking and machine-readable CSV exports exist. A scheduled automated MoPB feed, separate MoPB role.
10	Facility-Level Module	**Completed**	The platform contains 471 facilities across all 44 LGAs. Facility-user assignments, facility-scoped RLS, PHC validation, expenditure capture, LGA aggregation, facility ranking, and coverage reporting are implemented.
11	AI Analytics Module	**Substantial**	Rule-based analytics provide expenditure snapshots, execution variance, compliance signals, AOP linkage, off-pace/lagging identification, critical zero-release warnings, and reconciliation watchlists. A read-only conversational assistant and database tools are deployed.

No.	Contract deliverable	Status	Implementation evidence and assessment
12	Security Configuration	**Substantial**	HTTPS, authenticated sessions, server-only secrets, role and membership scoping, facility-level RLS, append-only BIR publication controls, audit triggers, constrained admin APIs, and protected storage policies are implemented.
13	UAT Report	**Evidence pending**	Automated tests and production verification are available.Signed UAT Report not available yet.
14	Pilot Deployment	**Substantial**	The production platform is live. The database contains 2,945 January-June monthly-tracking rows from 17 MDAs, exceeding the four-MDA numerical threshold in the excerpt. Formal confirmation that these are approved pilot submissions and the pilot acceptance record remain client-side sign-off items.

3. Implemented Platform Capabilities

Workbook and source-data reconciliation

- Mapped all workbook sheets, fields, formulas, lookups, dependent facility dropdowns, budget structure, and AOP logic.
- Recorded workbook defects and translated them into web validation and normalized database rules.
- Compared the Q2 workbook, published BIR, successive brief versions, and live database figures by MDA and component.
- Reconciled PHC programme classification so BIR Table 22 uses official NCOA programme codes and excludes PHCMB personnel in Section 3.

Funding and expenditure operations

- Separate funding and expenditure ledgers with fiscal-year public IDs, searchable lists, pagination, detail views, and controlled edits.
- Validated funding source, programme area, expenditure class/item, payment method, PHC, LGA, facility, AOP activity, and funding-allocation fields.
- Entry comments, attachments, audit history, data-quality warnings, notifications, and immutable quarter publication controls.
- Offline draft/reference caching, read-through data snapshots, queued operations, reconnect synchronization, and conflict/error handling.

Administration and planning

- Admin user creation, invitation, password reset, role assignment, MDA membership, and facility assignment.
- Reference-data maintenance for MDAs, programme areas, categories, funding sources, payment methods, LGAs, and facilities.
- Approved-budget and budget-line administration, AOP activities, funding partners, revenue lines, and monthly component tracking.
- Reference-value request workflow for controlled additions from operational users.

Dashboards, reporting, and exports

- MDA dashboards and statewide Admin Insights with fiscal year, quarter, MDA, programme, funding, category, PHC, LGA, and facility filters.
- BIR, audit/reconciliation, CSO public-finance, and budget/planning report templates with charts, tables, findings, and period context.
- Budget execution, funding mix, economic/administrative classification, NCOA programme objectives, PHC LGA/facility, AOP, and unlinked-spend analysis.

- Downloadable PDF reports and machine-readable CSV registers; report editorial/publishing metadata and append-only BIR versions.

4. Live Production Data and Coverage Evidence

The following counts were obtained through read-only SQL against the connected production Supabase project on 6 August 2026.

The counts establish that the database is populated and that the main reference, planning, and reporting domains are operational.

Production dataset	Count/value	Implementation relevance
MDA reference records	22	Includes the 21 budgeted health MDAs plus the additional reference/aggregate record used by the system.
Approved MDA budgets - FY 2026	21	Supports all contracted MDA dashboard scopes.
Programme areas	24	Operational programme dimension available to forms and dashboards.
Expenditure categories / funding sources	11 / 10	Controlled financial classifications populated in the database.
LGAs / facilities	44 / 471	Full Kano geographic and PHC facility reference coverage.
AOP activities - FY 2026	2,160	Activity-level planning data for linkage and variance analysis.
Approved budget lines - FY 2026	855	NCOA economic and programme detail for official BIR classification.
Funding / expenditure entries - FY 2026	65 / 267	Live ledgers supporting dashboards, reports, and audit registers.
Monthly tracking rows / reporting MDAs	2,945 / 17	January-June component returns and cross-source reconciliation coverage.
Monthly tracked amount	NGN 16,086,364,353.88	Aggregate value in the monthly-tracking source, subject to the application's reconciliation rules.
Public reporting views	9	Database-backed summaries for budget, funding, expenditure, programme, PHC, AOP, and exceptions.

PHC reporting reconciliation delivered

The latest implementation corrects Primary Healthcare BIR Table 22 to use the official NCOA programme classification instead of the application's operational programme-area grouping.

For Q2 2026, the report now reproduces the published non-personnel PHC envelope:

Measure	Amount
Approved budget	NGN 14,917,852,748.29
Q2 actual	NGN 2,395,610,606.91
YTD actual	NGN 5,646,059,506.91
Balance	NGN 9,271,793,241.38
Execution	Approximately 37.85%

5. Quality, Security, and Deployment Verification

Verification area	Evidence	Result
Automated tests	Vitest suite across reporting, validation, access, offline queue, database helpers, and UI routes.	**306 passed across 27 files**
Type safety	TypeScript no-emit validation and Next.js build-time type checking.	**Passed**
Production build	Next.js optimized production build, including static and dynamic routes and the service-worker bundle.	**Passed**

Verification area	Evidence	Result
Source control	GitHub main branch deployed from commit `ccab1e6`: "Add reporting reconciliation and in-app assistant."	**Committed and pushed**
Firebase rollout	Firebase App Hosting backend `budget-tracker` in project `budget-tracker-2fb26`, region `us-east4`.	**Successful**
Live HTTP checks	Production root resolved to `/mda`, and `/assistant` loaded from the hosted application.	**HTTP 200**
Assistant API	UI and read-only database tooling are deployed;	**Completed**

Implemented application security controls

- Supabase authentication with server-side user validation and protected authenticated routes.
- Roles for Admin, Viewer/Reviewer, MDA user, and facility user, with explicit capability checks.
- MDA memberships and facility assignments enforced in both application logic and Postgres row-level security.
- Audit triggers and mandatory reasons for privileged corrections; entry history, comments, and attachment records.
- Append-only quarterly BIR publications, deep database locks, controlled amendments, and archived legacy snapshots.
- Server-only service credentials, public publishable keys only in the browser, and HTTPS production hosting.

Appendix A: Evidence Register

Evidence	Location/identifier	Purpose
Workbook review	`docs/workbook-study.md`	Workbook structure, fields, logic, discrepancies, and platform resolutions.
Architecture decisions	`docs/adr/*`, `docs/database-model.md`, `docs/supabase-implementation.md`	Architecture, data model, security, deployment, and server conventions.
Executable database	`supabase/migrations/*` and `supabase/seeds/*`	Schema, RLS, triggers, reporting views, reference data, budgets, AOP, and Q2 data.
Application modules	`app/*`, `src/routes/*`, `src/components/reporting/*`, `src/lib/offline/*`	Operational forms, dashboards, reports, exports, offline behavior, and administration.
Automated verification	`src/test/*`, `npm test`, `npx tsc --noEmit`, `npm run build`	Regression, type, integration, and production-build evidence.
Deployed revision	Git commit `ccab1e6cabfcaee8ec92dce4bbf2eb5fc76e47b3`	Production deployment baseline used for this report.
Production hosting	`budget-tracker-2fb26` / Firebase App Hosting / `us-east4`	Successful rollout and live HTTPS endpoint.
Production database	Supabase project `znvcxidepemlmqdhavpl`	Read-only dataset counts and PHC reconciliation validation.
Company branding	https://dumbledoretech.com and public `/logo.svg`	Logo, company name, positioning statement, and contact details.

Links

- Production health finance tracker
- Deployed source revision on GitHub